

Lab07

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```
path <- "\\smb/ra281972/Downloads/ME315"
fname <- file.path(path, "database.sqlite3")

conn <- dbConnect(RSQLite::SQLite(), fname)
```

```
query_gpa_professores <- "
SELECT
  i.name AS professor,
  SUM(gd.a_count * 4.0 +
      gd.ab_count * 3.5 +
      gd.b_count * 3.0 +
      gd.bc_count * 2.5 +
      gd.c_count * 2.0 +
      gd.d_count * 1.0 +
      gd.f_count * 0.0) * 1.0 /
  SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count +
      gd.d_count + gd.f_count) AS gpa_medio,
  SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count +
      gd.d_count + gd.f_count) AS total_alunos
FROM instructors i
INNER JOIN teachings t ON i.id = t.instructor_id
INNER JOIN sections sec ON t.section_uuid = sec.uuid
INNER JOIN course_offerings co ON sec.course_offering_uuid = co.uuid
INNER JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
INNER JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
INNER JOIN subjects s ON sm.subject_code = s.code
WHERE s.abbreviation = 'STAT'
GROUP BY i.id, i.name
HAVING total_alunos > 0;
"

professores_gpa <- dbGetQuery(conn, query_gpa_professores)
```

```
prof_mais_dificil <- professores_gpa %>% arrange(gpa_medio) %>% slice(1)
prof_mais_facil <- professores_gpa %>% arrange(desc(gpa_medio)) %>% slice(1)
cat("Professor Mais Difícil:\n")
```

Professor Mais Difícil:

```
print(prof_mais_dificil)
```

```
      professor gpa_medio total_alunos
1 JAMES D KUELBS  2.607692           65
```

```
cat("\nProfessor Mais Fácil:\n")
```

Professor Mais Fácil:

```
print(prof_mais_facil)
```

```
      professor gpa_medio total_alunos
1 SRIKANTHMADHAVAN ARAVAMUTHAN           4           16
```

```
query_gpa_disciplinas <- "
SELECT
  c.name AS nome_disciplina,
  c.number AS numero_disciplina,
  SUM(gd.a_count * 4.0 +
    gd.ab_count * 3.5 +
    gd.b_count * 3.0 +
    gd.bc_count * 2.5 +
    gd.c_count * 2.0 +
    gd.d_count * 1.0 +
    gd.f_count * 0.0) * 1.0 /
  SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count +
    gd.d_count + gd.f_count) AS gpa_medio,
  SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count +
    gd.d_count + gd.f_count) AS total_alunos
```

```

FROM courses c
INNER JOIN course_offerings co ON c.uuid = co.course_uuid
INNER JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
INNER JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
INNER JOIN subjects s ON sm.subject_code = s.code
WHERE s.abbreviation = 'STAT'
GROUP BY c.uuid, c.name, c.number
HAVING total_alunos > 0;
"

disciplinas_gpa <- dbGetQuery(conn, query_gpa_disciplinas)

disc_mais_dificil <- disciplinas_gpa %>% arrange(gpa_medio) %>% slice(1)

disc_mais_facil <- disciplinas_gpa %>% arrange(desc(gpa_medio)) %>% slice(1)

cat("Disciplina Mais Difícil:\n")

```

Disciplina Mais Difícil:

```
print(disc_mais_dificil)
```

```

               nome_disciplina numero_disciplina gpa_medio
1 Introduction to the Theory of Probability          431  2.886374
  total_alunos
1          2231

```

```
cat("\nDisciplina Mais Fácil:\n")
```

Disciplina Mais Fácil:

```
print(disc_mais_facil)
```

```

               nome_disciplina numero_disciplina
1 Nonparametric Statistics and Machine Learning Methods          841
  gpa_medio total_alunos
1          4          24

```

```
dbDisconnect(conn)
```